Week 7

Topic: Greedy

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45. Jump Game II (Medium)

You are given a **0-indexed** array of integers nums of length n. You are initially positioned at index 0.

Each element nums[i] represents the maximum length of a forward jump from index i. In other words, if you are at index i, you can jump to any index (i + j) where:

- 0 <= j <= nums[i] and
- i + j < n

Return *the minimum number of jumps to reach index* n - 1. The test cases are generated such that you can reach index n - 1.

Example 1:

Input: nums = [2,3,1,1,4]

Output: 2

Explanation: The minimum number of jumps to reach the last index is 2. Jump 1 step from index 0 to 1, then 3 steps to the last index.

For detailed descriptions and constraints, please refer to Leetcode 45.

1383. Maximum Performance of a Team (Hard)

You are given two integers n and k and two integer arrays speed and efficiency both of length n. There are n engineers numbered from 1 to n. speed[i] and efficiency[i] represent the speed and efficiency of the ith engineer respectively.

Choose **at most** k different engineers out of the n engineers to form a team with the maximum **performance**.

The performance of a team is the sum of its engineers' speeds multiplied by the minimum efficiency among its engineers.

Return *the maximum performance of this team*. Since the answer can be a huge number, return it **modulo** $10^9 + 7$.

Example 1:

Input: n = 6, speed = [2,10,3,1,5,8], efficiency = [5,4,3,9,7,2], k = 2

Output: 60

Explanation:

We have the maximum performance of the team by selecting engineer 2 (with speed=10 and efficiency=4) and engineer 5 (with speed=5 and efficiency=7). That is, performance = (10 + 5) * min(4, 7) = 60.

For detailed descriptions and constraints, please refer to Leetcode 1383.

135. Candy (Hard)

There are n children standing in a line. Each child is assigned a rating value given in the integer array ratings.

You are giving candies to these children subjected to the following requirements:

- Each child must have at least one candy.
- Children with a higher rating get more candies than their neighbors.

Return *the minimum number of candies you need to have to distribute the candies to the children*.

Example 1:

Input: ratings = [1,0,2]

Output: 5

Explanation: You can allocate to the first, second and third child with 2, 1, 2 candies respectively.

For detailed descriptions and constraints, please refer to Leetcode 135.